

1. PROOF

Theorem 1.1 (Imbalance conjecture). *Let G be a finite simple graph such that $d_G(u) \neq d_G(v)$ for every edge uv . Then the multiset*

$$(|d_G(u) - d_G(v)|)_{uv \in E(G)}$$

is the degree multiset of a finite simple graph.

Proof. Write $d(v) = d_G(v)$, $D = \max_{v \in V(G)} d(v)$, and $\iota(uv) = |d(u) - d(v)|$. For $A \subseteq E(G)$ with $|A| = k$, put

$$T_A = \sum_{e \in E(G) \setminus A} \min\{k, \iota(e)\}.$$

We first prove

$$T_A \geq k(D - k)^+, \quad (1.1)$$

where $q^+ = \max\{q, 0\}$.

The claim is immediate when $k = 0$ or $k \geq D$. Suppose that $0 < k < D$, set $r = D - k$, and choose a vertex x of degree D . Let a be the number of edges of A incident with x . Of the $D - a$ edges at x outside A , choose r and let R be their opposite endpoints; let X be the opposite endpoints of the other such edges. Thus $|X| = k - a$.

For $z \in R$, maximality of $d(x)$ and the hypothesis on adjacent degrees give $d(z) < D$ and $\iota(xz) = D - d(z)$. Consequently,

$$\min\{k, \iota(xz)\} + (d(z) - r)^+ = k. \quad (1.2)$$

Indeed, if $d(z) \leq r$, the two terms are k and 0 ; otherwise they are $D - d(z)$ and $d(z) - r$.

Set $\rho = \sum_{z \in R} (d(z) - r)^+$. Call an edge an escape edge if it has one endpoint in R and the other outside $R \cup \{x\}$, and let F be the set of escape edges. For each $z \in R$, at most r possible neighbors of z lie in $R \cup \{x\}$. Since every escape edge has exactly one endpoint in R , summing over R gives

$$|F| \geq \rho. \quad (1.3)$$

By (1.2), the r chosen edges at x contribute $rk - \rho$ to T_A . No edge of F is incident with x , while exactly a edges of A are incident with x ; hence, by (1.3), at least $(\rho - (k - a))^+$ escape edges lie outside A . Each contributes at least one to T_A , as does each of the $k - a$ edges from x to X . These three groups of edges are pairwise disjoint, so

$$T_A \geq rk - \rho + (k - a) + (\rho - (k - a))^+ \geq rk.$$

As $r = D - k$, this proves (1.1).

Every edge has imbalance at most $D - 1$: both endpoints have positive degree at most D . It follows that every $A \subseteq E(G)$ of size k satisfies

$$\sum_{e \in A} \iota(e) \leq k(k - 1) + \sum_{e \in E(G) \setminus A} \min\{k, \iota(e)\}. \quad (1.4)$$

If $D \leq k$, the left side is at most $k(D-1) \leq k(k-1)$. If $k < D$, it is at most

$$k(D-1) = k(k-1) + k(D-k),$$

and (1.1) supplies the last term in (1.4).

The sum of all imbalances is even. Since $|p - q| \equiv p + q$ modulo two, summing first over edges and then over incidences gives

$$\sum_{uv \in E(G)} |d(u) - d(v)| \equiv \sum_{uv \in E(G)} (d(u) + d(v)) = \sum_{v \in V(G)} d(v)^2.$$

But $d(v)^2 \equiv d(v)$ modulo two, and

$$\sum_{v \in V(G)} d(v) = 2|E(G)|,$$

so the imbalance sum is even.

If $E(G)$ is empty, the conclusion is immediate. Otherwise, write the imbalances in nonincreasing order as $s_1 \geq \dots \geq s_m$, where $m = |E(G)|$. For $1 \leq k \leq m$, choose A to consist of k edges carrying $s_1, \dots, s_k$. Then (1.4) says

$$\sum_{i=1}^k s_i \leq k(k-1) + \sum_{i=k+1}^m \min\{k, s_i\}.$$

Together with the even sum just proved, these are precisely the Erdős–Gallai conditions. Their sufficiency shows that $(s_1, \dots, s_m)$ is graphic. $\square$